→ aaAbb (by A → aAb)

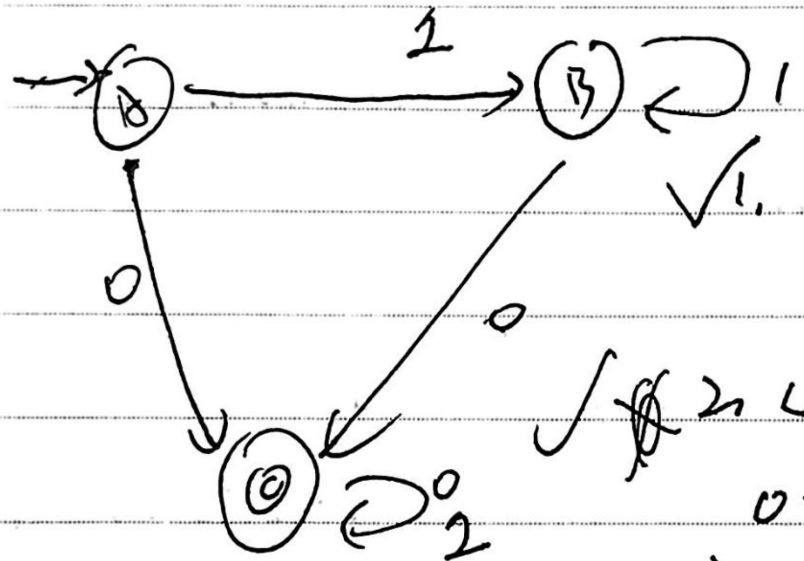
→ aaaAbbb

→ aaabbb (by A → ε)

→ $a^3 b^3$ ⇒ $a^n b^n$

66 Regular Languages and Finite Automata

Problem - 1



which False

✓1. Complement of L(A) is context free

✓ 2. $L(A) = L((11^*0+0)(0+1)^*0^*1^*)$

3. For the Language accepted by A, A is the minimal DFA

4. A accepts all strings

over {0,1} of length at least 2. X

	A	B	C
A			
B	O		
C	✓	✓	

$(11^*0+0)(0+1)^*0^*1^*$

$01^*(0+1)^*$ ✓

$\delta(A,0)=C$
$\delta(B,0)=C$ ⇒ Unmark

0 Equ: {A,B}, {C}
1 Equ: {A,B}, {C}
→ A,B can be combined ✓

↑ use this method

(A,B) $\xrightarrow{0}$ (C,C) $\xrightarrow{1}$ (B,B)

~~Exits~~ ~~Exits~~

it's the minimal DFA

$0(\epsilon+\underline{11^*})$

$L(A)=$

$A=\epsilon$

$B = A1 + B1 = A1\cdot 1^* = \cancel{A\cdot 1^+} = 11^*$

$C = A0 + B0 + C0 + C1$

$C = 0 + 11^*0 + C0 + C1$

$= \cancel{0+11^*0} = 0 + 11^*0(0+1)^*$

$= \cancel{0\,11^* + C(0+1)} = 11^*0 + 0(0+1)^*$

$= \cancel{0\,11^*(0+1)^*}$

$= 01^* + C(0+1)^*$

$= 01^*(0+1)^*$

(67) problem 2,

$L_1 = \phi$, $L_2 = \{a\}$, $L_1 L_2^* \cup L_1^*$?

(A) ϵ

(B) ϕ

(C) a^*

(D) $\{\epsilon, a\}$

ϕ $\quad L_1^* = \epsilon$

$\Rightarrow \phi \cup \epsilon = \epsilon$

$\Rightarrow A$

(68) Problem 3,

$\{0, 1\}$, every substring of 3 symbols has at most two zeros. ~~&~~ or less than 3 length

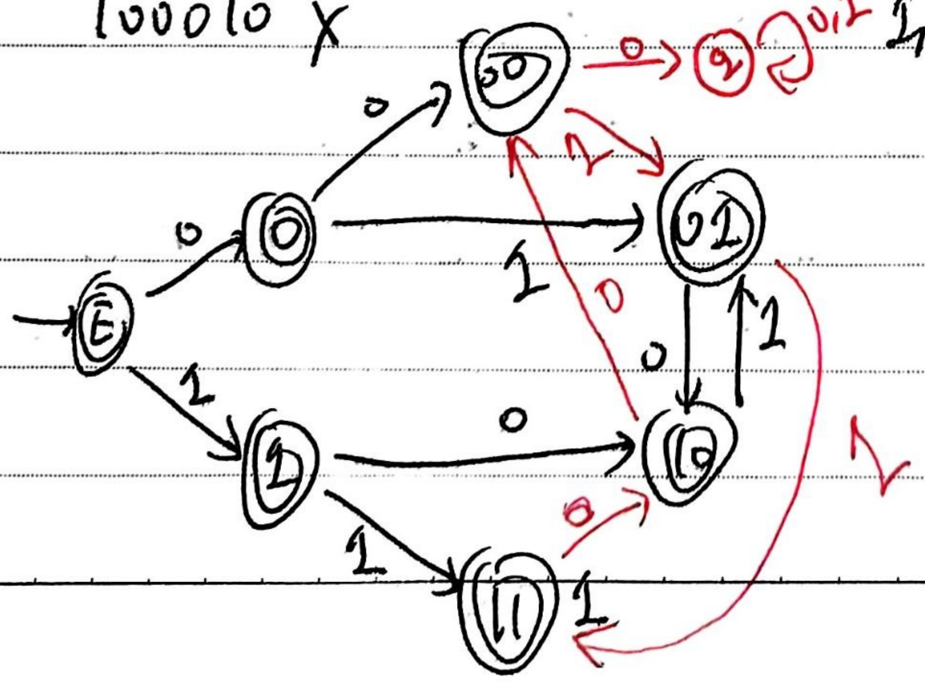
00 / 110, 011, 001 ✓

100010 ✗

11 ✓

1, 0, 01, 00, ~~100, 101~~

~~111~~ ✓



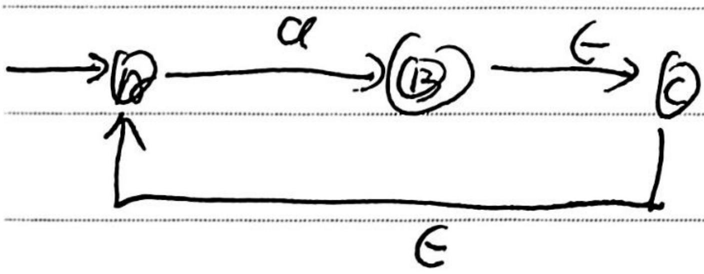
(69) Problem 4

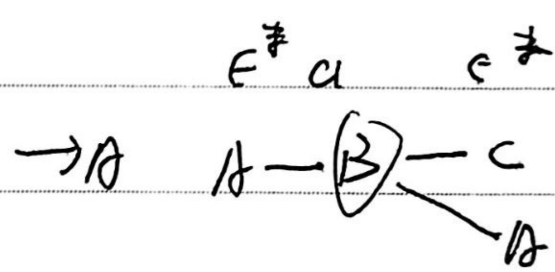
Which one of the following languages over the alphabet $\{0,1\}$ is described by the regular expression: $(0+1)^*0(0+1)^*0(0+1)^*$?

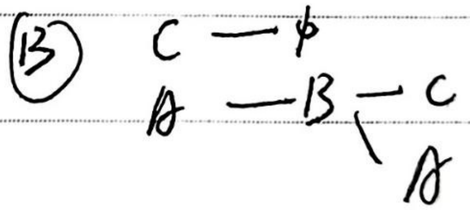
(C) The set of all strings containing at least two 0's.

(70) Problem 5.

What is the complement (oppsite) of the languages accepted by the NFA shown below?







(A) ϕ $\cancel{=}$

(B) $\{\epsilon\}$ ✓

(C) a^*

(D) $\{a, \epsilon\}$

~~(C)~~

$\Rightarrow$ $\cancel{a^*}$ a^+

So complement is ϵ

Mo | Tu | We | Th | Fr | Sa | Su

Memo No. ______

Date

(21) Problem 6,

$L = \{aa, ab, baa\}$ which in L^*?

1) abaabaaabaa ✓

2) aaaabaaaa ✓

3) baaaaabaaaab X

4) baaaaabaa ✓

1), 2), 4)